

Co-Ordinate Geometry

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Part I

Straight Line

Chapter 1

The Straight Line

1.0.0.1 Single Answer Type

Example 1 : i) If a, b, c form a G.P. with common ratio r , the product of the y-coordinates of the points of intersection of the line $ax+by+c=0$ and the curve $y^2 + 2x = 0$. ii) Will the intersection coordinates be real if we choose the intersecting curve to be $y^2 = 2x$.

- i)
- a) $-r^2$
- b) $-2r$
- c) $-2r^2$
- d) $r/2$
- ii)
- a) Yes
- b) No
- c) Can't say
- d) None of these.

{ Hint : Answer i) c) ii) b)

Solution : i) As a, b, c are in GP, put $b = ar, c = ar^2$

This gives, $ax + ary + ar^2 = 0$

i.e. $x + ry + r^2 = 0$

putting $x = -\frac{y^2}{2}$, we get

$$-y^2 + 2ry + 2r^2 = 0$$

Sum of roots $= y_1 + y_2 = 2r$

Product of roots $= y_1 y_2 = -2r^2$

$$|y_1 - y_2| = \sqrt{(y_1 + y_2)^2 - 4y_1 y_2} = \sqrt{4r^2 + 8r^2} = \text{Real}$$

Hence, the equation will have real y-coordinates . This can be figured out by drawing the graph of the solution.

ii) In this case, $x = \frac{y^2}{2}$

$$y^2 + 2ry + 2r^2 = 0$$

Sum of roots $= y_1 + y_2 = -2r$

Product of roots $= y_1 y_2 = 2r^2$

$$|y_1 - y_2| = \sqrt{(y_1 + y_2)^2 - 4y_1 y_2} = \sqrt{4r^2 - 8r^2} = \text{Imaginary}$$

Hence, the equation will not have real roots.

}

Example 2: Orthocentre of the triangle with vertices $(0,0)$, $(4,0)$, $(0,3)$ is

- a) $(3/4,4)$
- b) $(3,4)$
- c) $(0,0)$
- d) None of these

{ Hint : By inspection c) (0,0) is the requisite answer . It can be further generalized that the orthocenter of the triangle formed by lines $x=0, y=0$ and any other third line is indeed (0,0). }

Example 3 : A straight line through the origin divides the parallel lines $\sqrt[3]{3}x + 5y + 8 = 0$ and $3\sqrt[3]{3}x + 15y = 10$ in the ratio

- a) 8/5
- b) 5/8
- c) 2/9
- d) 0.3

{Hint : Let the line be $y=mx$

The point of intersection with the first line would be $\left(\frac{-8}{\sqrt[3]{3} + 5m}, \frac{-8m}{\sqrt[3]{3} + 5m} \right)$

Similarly, the point of intersection with the second line would be $\left(\frac{10/3}{\sqrt[3]{3} + 5m}, \frac{10m/3}{\sqrt[3]{3} + 5m} \right)$

Using section formula, the ratio in which (0,0) divides these the line segment joining these two points would be $8 \times 3 / 10 = 8/5$ }

Example 4 : The sum of distances of a point from two perpendicular lines is $2\sqrt{2}$. The are enclosed by the locus of these points is.

- a) 16
- b) 12
- c) 32
- d) 8

{ Hint : Answer 16 a)

Let's take the two lines to be $x=0$ and $y=0$

$\Rightarrow$ The equation of locus is $|x| + |y| = 2\sqrt{2}$

$\Rightarrow$ The equation is basically 4 line segments joining the vertices as a square as for the values $x > 2\sqrt{2}$ or $y > 2\sqrt{2}$, the equality can't hold for real values of x and y .

So, the area diagonals are of length $4\sqrt{2}$ each. That implies, the area of square is 16 Units. }

1.0.0.2 Subjective Problems

Example : The number of integral points (integral point means both the coordinates should be integer) that lie exactly in the interior of the triangle with vertices (0,0),(0,m),(n,0) where m,n are both +ve integers with $m > n$. {Subjective}

{ Hint : By inspection, moving down from the topmost point, at the line $y=m-1$, the farthestmost point inside the triangle is the left hand limit of n/m . On this line, the integral points is Zero or $\left[\frac{n}{m} \right]$.

On the line $y=m-2$, the number of points is $\left[\frac{2n}{m} \right]$

Proceeding in this way, on the last line with integral points, the equation is $y=1$, and the number of points is $\left[\frac{(m-2)n}{m} \right]$

Hence , the total number of such points is $\left[\frac{n}{m} \right] + \left[\frac{2n}{m} \right] + \dots + \left[\frac{(m-2)n}{m} \right]$ }

Part II

Ellipse

Chapter 2

JEE Mains Problems

Q1: Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a < b$, pass through the point $(4, 3)$ and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

- (A) $\frac{4\sqrt{5}}{3}$
- (B) $2\sqrt{5}$
- (C) $\frac{7\sqrt{5}}{3}$
- (D) $\frac{8\sqrt{5}}{3}$

{Solution : <https://www.youtube.com/watch?v=PbL0Q3oPqWI>}